MF 281: Probability Theory

Week 1: Sets, functions and counting

The language probability is written in

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Lecture notes for students of MF 281 Probability Theory.

Week 1: Sets, functions and counting

Learning Objectives

By the end of this week you should be able to:

- write events as sets, and translate between English and set notation;
- compute unions, intersections, complements and differences, and draw them;
- state and prove De Morgan's laws, and use them to simplify an event;
- recognise a partition, and say why partitions matter for probability;
- form Cartesian products and count their elements;
- use the counting principle, permutations, combinations and partitions to count outcomes without listing them;
- apply inclusion and exclusion to counts that overlap.

1.1 Why a probability course begins here

Probability assigns numbers to events. Before any number can be assigned, the event has to be described exactly, and the language for that is set theory. "At least one of the three pumps fails" is an English sentence; $A_1 \cup A_2 \cup A_3$ is the same statement in a form you can compute with.

The second half of this week is counting. For an experiment with finitely many equally likely outcomes, probability reduces to

$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } \Omega},$$

so the whole problem becomes a counting problem. Most mistakes in elementary probability are counting mistakes, not probability mistakes.

1.2 Sets

Definition 1.1 (Set)**.** A collection of distinct objects, called its elements. We write $x \in A$ when x belongs to A, and $x \notin A$ when it does not.

A set may be listed, $A = \{2, 4, 6\}$, or described, $A = \{x \in \mathbb{N} : x \leq 6 \text{ and } x \text{ is even}\}$. Order and repetition carry no meaning: $\{2, 4, 6\}$, $\{6, 4, 2\}$ and $\{2, 2, 4, 6\}$ are the same set.

Sets you will meet constantly: the empty set $\varnothing$, the natural numbers $\mathbb{N}$, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$ and the reals $\mathbb{R}$. In probability, the set of everything that could happen is the **sample space** Ω, and an **event** is a subset of Ω.

$A \subseteq B$ means every element of A is in B. Two sets are equal exactly when $A \subseteq B$ and $B \subseteq A$, which is how most set identities are proved.

1.2.1 Operations

Name	Symbol	Elements	In words, for events
Union	$A \cup B$	in A or B or both	at least one occurs
Intersection	$A \cap B$	in both	both occur
Complement	A^c	in Ω, not in A	A does not occur
Difference	$A \setminus B$	in A, not in B	A occurs and B does not
Symmetric difference	$A \triangle B$	in exactly one	exactly one of them occurs

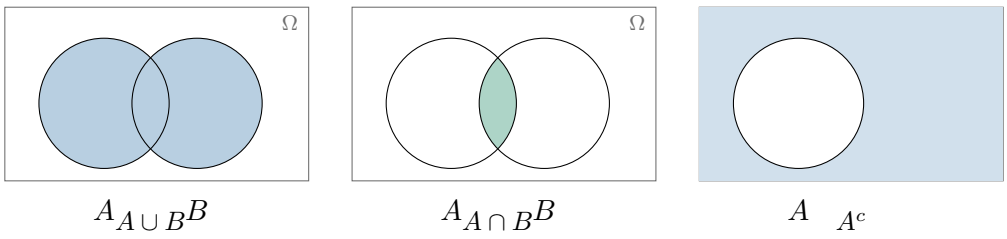


Figure 1.1: Union, intersection and complement. A Venn diagram is a picture of a logical statement, and it is the fastest way to check one.

Example 1: one die, five operations

Roll a fair die once, so $\Omega = \{1,2,3,4,5,6\}$. Let A be "even" and B be "at least 4", that is $A = \{2,4,6\}$ and $B = \{4,5,6\}$.

Solution

$$A \cup B = \{2,4,5,6\} \quad A \cap B = \{4,6\} \quad A^c = \{1,3,5\}$$
$$A \setminus B = \{2\} \qquad\qquad A \triangle B = \{2,5\}$$

Read each one back in English as a check: $A \setminus B = \{2\}$ says the only way to roll an even number that is not at least 4 is to roll a 2, which is plainly right.

1.2.2 The laws, and one proof

Commutative	$A \cup B = B \cup A, \quad A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
De Morgan	$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c$

De Morgan's laws are the ones you will use most, because "not at least one" is usually easier to count than "at least one".

Theorem 1.1 (De Morgan)**.** $(A \cup B)^c = A^c \cap B^c$.

Proof. Take $x \in (A \cup B)^c$. Then $x \notin A \cup B$, so x is in neither A nor B; hence $x \in A^c$ and $x \in B^c$, so $x \in A^c \cap B^c$. This gives $(A \cup B)^c \subseteq A^c \cap B^c$.

Now take $x \in A^c \cap B^c$. Then $x \notin A$ and $x \notin B$, so x is not in $A \cup B$, that is $x \in (A \cup B)^c$. This gives the reverse inclusion, and the two together give equality. □

That two-step pattern, "take an element of the left, show it lies in the right, then the other way round", proves almost every set identity you will be asked for. Learn it once this week and the rest of the term costs you nothing.

Example 2: De Morgan on the die

With $A = \{2,4,6\}$ and $B = \{4,5,6\}$ from Example 1:

$$A \cup B = \{2,4,5,6\} \Rightarrow (A \cup B)^c = \{1,3\},$$

$$A^c = \{1,3,5\}, \quad B^c = \{1,2,3\} \Rightarrow A^c \cap B^c = \{1,3\}.\checkmark$$

Why it matters

In probability this becomes $P(\text{at least one}) = 1 - P(\text{none})$, which turns a hard union into an easy intersection. Nearly every "at least one" question in this course is solved that way.

1.2.3 Disjointness and partitions

Sets are **disjoint** when $A \cap B = \varnothing$: they cannot both happen.

Definition 1.2 (Partition)**.** Sets $A_1, A_2, \ldots, A_n$ partition Ω if they are pairwise disjoint and $\bigcup_{i=1}^{n} A_i = \Omega$: exactly one of them happens.

Partitions are the backbone of the total probability theorem and Bayes' rule, which we reach in Week 4. Whenever you see "split into cases", the cases must form a partition, or you will double count or miss outcomes.

1.3 Cartesian products

Definition 1.3 (Cartesian product)**.** $A \times B = \{(a, b) : a \in A,\ b \in B\}$, the set of ordered pairs.

Order matters here, unlike in a set: $(2, 4) \neq (4, 2)$. If A has m elements and B has n, then $A \times B$ has mn, which is the counting principle in disguise.

Example 3: a die and a card suit

Roll a die and draw a suit: $\Omega = \{1, \ldots, 6\} \times \{\heartsuit, \diamondsuit, \clubsuit, \spadesuit\}$, so $|\Omega| = 6 \times 4 = 24$ equally likely outcomes, from $(1, \heartsuit)$ to $(6, \spadesuit)$.

Solution

Products are how sample spaces for repeated or combined experiments are built. Three tosses of a coin live in $\{H, T\}^3$, with $2^3 = 8$ elements; n tosses give 2^n, and that doubling is why exhaustive listing stops being possible almost immediately.

1.4 Functions, briefly

Definition 1.4 (Function)**.** $f : A \to B$ assigns to each $a \in A$ exactly one $f(a) \in B$. A is the domain, B the codomain, and $\{f(a) : a \in A\}$ the image.

A function is **injective** if different inputs give different outputs, **surjective** if every element of B is hit, and **bijective** if both, in which case it pairs up the two sets exactly.

Two uses in this course. First, a **random variable** is nothing more exotic than a function from Ω to $\mathbb{R}$: it attaches a number to each outcome, and we study it from Week 5. Second, the **indicator** of an event,

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

turns statements about sets into arithmetic, which is often the neatest way to prove something.

Counting is itself a bijection argument. Saying "there are $n!$ orderings" means there is a one-to-one pairing between orderings and the numbers $1, \ldots, n!$. When a counting problem resists, look for a bijection with something you can already count.

1.5 Counting

1.5.1 The counting principle

If a procedure has k stages, with n_1 choices at the first, n_2 at the second and so on, the whole procedure has $n_1 n_2 \cdots n_k$ outcomes, provided the number of choices at each stage does not depend on what went before.

Example 4: number plates

A plate is three letters followed by three digits, repetition allowed. How many?

Solution

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 = 26^3\,10^3 = 17{,}576{,}000.$$

Now forbid repeated letters and repeated digits:

$$26 \times 25 \times 24 \times 10 \times 9 \times 8 = 15{,}600 \times 720 = 11{,}232{,}000,$$

about 36% fewer. The stages still multiply; only the counts change.

1.5.2 Permutations

An arrangement of all n distinct objects in order can be made in

$$n! = n(n-1)(n-2)\cdots 1$$

ways, with $0! = 1$ by convention. Choosing and arranging k of the n gives the k-permutation count

$$P(n,k) = \frac{n!}{(n-k)!}.$$

Example 5: shelves and prizes

(a) Seven distinct books on a shelf: $7! = 5040$ arrangements.
(b) Ten students compete for gold, silver and bronze, one each:

$$P(10,3) = \frac{10!}{7!} = 10 \times 9 \times 8 = 720.$$

Solution

In (b) order matters, because gold and silver are different outcomes. If the three simply formed a team, order would not matter, and the answer would be the combination in Example 6 instead. Deciding whether order matters is the whole art of counting.

1.5.3 Combinations

Choosing k from n without regard to order:

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$

The division by $k!$ removes the orderings of the chosen set, which is the same bijection idea as before.

Example 6: committees

(a) A committee of 4 from 11 students: $\binom{11}{4} = 330$.
(b) A committee of 4 from 6 men and 5 women, which must have exactly 2 of each:

$$\binom{6}{2}\binom{5}{2} = 15 \times 10 = 150.$$

(c) A committee of 3 from 3 women and 5 men, with *at least* one woman:

$$\binom{8}{3} - \binom{5}{3} = 56 - 10 = 46.$$

Solution

Part (c) shows the complement trick again. Counting "at least one" directly means adding the cases of one, two and three women ($3\cdot 10+3\cdot 5+1=46$), which works but is slower and easier to get wrong.

1.5.4 Partitions into groups

Splitting n distinct objects into labelled groups of sizes $n_1,\dots,n_r$ with $\sum n_i = n$:

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1!\, n_2! \cdots n_r!}.$$

Example 7: project groups

Ten students are split into a group of 4 and two groups of 3.

Solution

If the groups are labelled (group A does project 1, and so on):

$$\frac{10!}{4!\,3!\,3!} = 4200.$$

If the two groups of three are interchangeable, because the labels mean nothing, each arrangement has been counted twice, so divide by $2! = 2$:

$$\frac{4200}{2} = 2100.$$

The lesson

Labelled or not is not a detail. It is a factor of two here, and a factor of $r!$ when r groups are interchangeable. Decide before you compute, and say which you assumed.

1.5.5 Inclusion and exclusion

For overlapping sets, adding sizes double counts the overlap:

$$|A\cup B| = |A| + |B| - |A\cap B|,$$
$$|A\cup B\cup C| = |A| + |B| + |C| - |A\cap B| - |A\cap C| - |B\cap C| + |A\cap B\cap C|.$$

Example 8: who uses what

Of 120 students, 70 use R, 55 use Python and 30 use both.

Solution

$$|R\cup P| = 70 + 55 - 30 = 95,$$

so 95 use at least one and $120-95 = 25$ use neither. Note that the careless answer, $70+55 = 125$, exceeds the class size, which is the signal that an overlap was double counted.
With a third tool, say 40 using MATLAB, 25 using R and MATLAB, 20 using Python and MATLAB, and 15 using all three:

$$70 + 55 + 40 - 30 - 25 - 20 + 15 = 105.$$

1.6 Back to probability

With Ω finite and all outcomes equally likely,

$$P(A) = \frac{|A|}{|\Omega|},$$

so everything above is immediately a probability statement. Two from this chapter:

- Example 6(c) gives $P(\text{at least one woman}) = 46/56 = 0.82$.
- Example 8 gives $P(\text{a student chosen at random uses neither}) = 25/120 = 0.21$.

The equally likely assumption is doing real work there, and it is an assumption about the world, not a mathematical fact. Week 2 replaces it with the axioms, which let us handle the cases where outcomes are not equally likely at all.

1.7 A checklist for counting problems

1. What exactly is one outcome? Write one down.
2. Does order matter? Permutation if yes, combination if no.
3. Is repetition allowed?
4. Are the objects distinguishable? Are the groups labelled?
5. Would the complement be easier to count?
6. Does the answer pass a sanity check: not negative, not larger than the total, not smaller than a part?

1.8 Glossary

Ω	Sample space: the set of all possible outcomes.
Event	A subset of Ω.
A^c	Complement: the outcomes in Ω not in A.
Disjoint	$A \cap B = \varnothing$; the two cannot both occur.
Partition	Pairwise disjoint sets whose union is Ω.
$A \times B$	Cartesian product: ordered pairs, $\lvert A \times B\rvert = \lvert A\rvert\,\lvert B\rvert$.
$n!$	Arrangements of n distinct objects.
$P(n,k)$	Ordered selections of k from n: $n!/(n-k)!$.
$\binom{n}{k}$	Unordered selections: $n!/(k!(n-k)!)$.
Inclusion–exclusion	Correction for double counting overlaps.

1.9 Summary

Week 1 in six lines

- Events are sets; set notation is how a question becomes computable.
- De Morgan turns "at least one" into "not none", which is usually easier.
- A partition is a split into cases that is neither overlapping nor incomplete.
- Cartesian products build the sample spaces of combined experiments.
- Order matters means permutation; order does not means combination.
- With equally likely outcomes, probability is a counting problem.

Exercises

1. With $\Omega = \{1, \ldots, 10\}$, A the even numbers, B the primes and C the multiples of 3, list $A \cap B$, $A \cup C$, B^c, $A \setminus C$ and $B \triangle C$.

2. Prove the second De Morgan law, $(A \cap B)^c = A^c \cup B^c$, in the two-step style used above.

3. Three pumps operate independently. Write in set notation: all three fail; none fails; exactly one fails; at least two fail.

4. Do A, $A^c \cap B$ and $A^c \cap B^c$ partition Ω? Justify your answer.

5. A PIN is four digits with no digit repeated. How many are there? What fraction of all four-digit strings is that?

6. A password is 8 characters from the 26 lower-case letters and 10 digits, repetition allowed. How many? If it must contain at least one digit, how many?

7. Nine distinct components are placed in a row. In how many orders? In how many if three particular components must be adjacent?

8. From 7 engineers and 4 geologists, choose a team of 5 with at least 2 geologists. How many teams?

9. Twelve students are split into three labelled groups of 4. How many ways? And if the groups are unlabelled?

10. In a class of 90, 48 passed statistics, 52 passed programming and 21 passed neither. How many passed both?

11. A fair coin is tossed six times. How many outcome sequences are there, and how many contain exactly four heads? Hence find the probability of exactly four heads.

12. Explain, in two sentences and without formulas, why $\binom{n}{k} = \binom{n}{n-k}$.

Answers to the odd-numbered exercises

1. $A = \{2,4,6,8,10\}$, $B = \{2,3,5,7\}$, $C = \{3,6,9\}$. Then $A \cap B = \{2\}$; $A \cup C = \{2,3,4,6,8,9,10\}$; $B^c = \{1,4,6,8,9,10\}$; $A \setminus C = \{2,4,8,10\}$; $B \triangle C = \{2,5,6,7,9\}$.

3. With F_i the event that pump i fails: all three, $F_1 \cap F_2 \cap F_3$; none, $F_1^c \cap F_2^c \cap F_3^c$; exactly one, $(F_1 \cap F_2^c \cap F_3^c) \cup (F_1^c \cap F_2 \cap F_3^c) \cup (F_1^c \cap F_2^c \cap F_3)$; at least two, $(F_1 \cap F_2) \cup (F_1 \cap F_3) \cup (F_2 \cap F_3)$.

5. $P(10,4) = 10 \times 9 \times 8 \times 7 = 5040$. All four-digit strings number $10^4 = 10{,}000$, so the fraction is 0.504.

7. $9! = 362{,}880$. Treating the three as one block gives $7!$ arrangements of the blocks and $3!$ within, so $7! \times 3! = 5040 \times 6 = 30{,}240$.

9. Labelled: $12!/(4!\,4!\,4!) = 34{,}650$. Unlabelled: divide by $3! = 6$, giving 5775.

11. $2^6 = 64$ sequences, of which $\binom{6}{4} = 15$ have exactly four heads, so the probability is $15/64 = 0.234$.

Before next week

Work exercises 1 to 6 before the next class; we begin with them. Week 2 states the axioms of probability and derives the basic laws from them, so that we are no longer restricted to outcomes that happen to be equally likely.